Probability, Statistics and Random Processes

Third Edition

T Veerarajan
Professor of Mathematics and Head
Science and Humanities Department
Sree Sowdambika College of Engineering
Aruppukottai, Tamil Nadu

Tata McGraw-Hill Publishing Company Limited
NEW DELHI

McGraw-Hill Offices

New Delhi New York St Louis San Francisco Auckland Bogotá Caracas
Kuala Lumpur Lisbon London Madrid Mexico City Milan Montreal
San Juan Santiago Singapore Sydney Tokyo Toronto

Tata McGraw-Hill

Published by the Tata McGraw-Hill Publishing Company Limited,
7 West Patel Nagar, New Delhi 110 008.

Copyright © 2008, 2006, by Tata McGraw-Hill Publishing Company Limited.
No part of this publication may be reproduced or distributed in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise or stored in a database or retrieval system without the prior written permission of the publishers. The program listing (if any) may be entered, stored and executed in a computer system, but they may not be reproduced for publication.

This edition can be exported from India only by the publishers,
Tata McGraw-Hill Publishing Company Limited.

ISBN 13: 978-0-07-066925-3
ISBN 10: 0-07-066925-2

Managing Director: *Ajay Shukla*

General Manager: Publishing—SEM & Tech Ed: *Vibha Mahajan*
Sponsoring Editor: *Shukti Mukherjee*
Editorial Executive: *Suman Sen*
Executive—Editorial Services: *Sohini Mukherjee*
Senior Production Assistant: *Stephen Manuvel*

General Manager: Marketing—Higher Education & School: *Michael J Cruz*
Product Manager: SEM & Tech Ed: *Biju Ganesan*

Controller—Production: *Rajender P Ghansela*
Asst. General Manager—Production: *B L Dogra*

Information contained in this work has been obtained by Tata McGraw-Hill, from sources believed to be reliable. However, neither Tata McGraw-Hill nor its authors guarantee the accuracy or completeness of any information published herein, and neither Tata McGraw-Hill nor its authors shall be responsible for any errors, omissions, or damages arising out of use of this information. This work is published with the understanding that Tata McGraw-Hill and its authors are supplying information but are not attempting to render engineering or other professional services. If such services are required, the assistance of an appropriate professional should be sought.

Published by the Tata McGraw-Hill Publishing Company Limited, 7 West Patel Nagar, New Delhi 110 008, typeset at Bukprint India, B-180A, Guru Nanak Pura, Laxmi Nagar-110 092 and printed at Adarsh Printers, C-50-51, Mohan Park, Naveen Shahdara, Delhi 110 032

Cover: SDR Printers

RCLACRQXRBLRA

The McGraw·Hill Companies

HYPOTHESIS TESTING

Introduction

Hypothesis testing is a statistical method used to determine whether a statement about a population parameter is likely to be true. It uses **sample data** to decide whether to accept or reject a hypothesis regarding the population.

A hypothesis must be:

- clear
- precise
- specific

It should clearly express the relationship between variables so that the claim can be tested statistically.

Statistical Hypothesis

A **statistical hypothesis** is a conjecture about a population parameter.

Two hypotheses are generally considered.

Null Hypothesis

The **null hypothesis** is denoted by

$$H_0$$

It states that **no significant difference exists** between a population parameter and a specified value.

Example

$$H_0 : \mu = 50$$

Alternative Hypothesis

The **alternative hypothesis** is denoted by

$$H_1$$

It contradicts the null hypothesis and indicates that **a significant difference exists**.

Example

$$H_1 : \mu \neq 50$$

Types of Tests

Depending on the alternative hypothesis, statistical tests are classified as follows.

Two-Tailed Test

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0$$

Both sides of the distribution are considered.

Right-Tailed Test

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu > \mu_0$$

Only the **right side of the distribution** is considered.

Left-Tailed Test

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu < \mu_0$$

Only the **left side of the distribution** is considered.

Errors in Hypothesis Testing

Two types of errors may occur.

Type I Error

Rejecting the null hypothesis when it is actually true.

Probability of Type I error

$$\alpha$$

This is called the **level of significance**.

Typical values

$$\alpha = 0.01, 0.05, 0.10$$

Type II Error

Accepting the null hypothesis when it is actually false.

Probability of Type II error

$$\beta$$

Decision Table

Reality	Decision Accept H_0	Decision Reject H_0
H_0 true	Correct decision	Type I error
H_0 false	Type II error	Correct decision

Z-Test (Large Sample Test)

When

$$n \geq 30$$

the sample is considered **large**.

Z-test is used when

- population standard deviation is known
- sample size is large
- distribution is normal or approximately normal

Critical Region

The **critical region** is the region of the sampling distribution where the null hypothesis is rejected.

The remaining region is the **acceptance region**.

Critical Values Table

Test Type	α = 1%	α = 2%	α = 5%	α = 10%
Two-tailed	2.58	2.33	1.96	1.645
Right-tailed	2.33	2.055	1.645	1.28
Left-tailed	−2.33	−2.055	−1.645	−1.28

Procedure for Hypothesis Testing

1. State the null and alternative hypotheses.
2. Determine the type of test.
3. Select the level of significance.
4. Calculate the test statistic.
5. Compare with the critical value.
6. Draw conclusion.

Test of Significance for Mean (Large Sample)

Let

$$x_1, x_2, \ldots, x_n$$

be a random sample.

Sample mean

$$\bar{x}$$

Population mean

$$\mu$$

Standard deviation

$$\sigma$$

Test statistic

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

If population standard deviation is unknown

$$Z = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Problem 1

Heights of college students are normally distributed with

$$\sigma = 6$$

Sample size

$$n = 100$$

Sample mean

$$\bar{x} = 158$$

Test

$$H_0 : \mu = 160$$

$$Z = \frac{158 - 160}{6 / \sqrt{100}}$$

$$Z = -3.333$$

Since

$$|Z| > 1.96$$

Reject H_0.

Problem 2

Sample size

$$n = 400$$

Population standard deviation

$$\sigma = 10$$

Sample mean

$$\bar{x} = 40$$

Population mean

$$\mu = 38$$

$$Z = \frac{40 - 38}{10/\sqrt{400}}$$

$$Z = 4$$

Since

$$4 > 1.96$$

Reject H_0.

Confidence interval

$$40 \pm 1.96 \frac{10}{\sqrt{400}}$$

$$(39.02, 40.98)$$

Problem 3

Production process mean

$$\mu = 50$$

Standard deviation

$$\sigma = 2.5$$

Sample size

$$n = 12$$

Sample mean

$$\bar{x} = 46.5$$

Test

$$H_1 : \mu < 50$$

$$Z = \frac{46.5 - 50}{2.5/\sqrt{12}}$$

$$Z = -4.854$$

Critical value

$$-1.645$$

Since

$$Z < -1.645$$

Reject H_0.

Problem 4

Sample size

$$n = 900$$

Sample mean

$$\bar{x} = 3.4$$

Population mean

$$\mu = 3.25$$

Standard deviation

$$\sigma = 2.61$$

$$Z = \frac{3.4 - 3.25}{2.61/\sqrt{900}}$$

$$Z = 1.724$$

Since

$$1.724 < 1.96$$

Accept H_0.

Difference Between Two Sample Means

Test statistic

$$Z = \frac{\bar{x}_1 - \bar{x}_2 - \delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Problem

Mean heights

Place	Mean	SD	n
A	68.50	2.5	1200
B	68.58	3.0	1500

$$Z = -0.756$$

Since

$$|Z| < 1.96$$

Accept H_0.

Example

Two large samples

$$n_1 = 1000, \quad n_2 = 2000$$

Means

$$67.5, 68$$

Population SD

$$2.5$$

$$Z = -5.16$$

Since

$$|Z| > 1.96$$

Reject H_0.

Example

English men vs Americans

$$n_1 = 6400, \quad n_2 = 1600$$

Means

$$170, \quad 172$$

SD

$$6.4, \quad 6.3$$

$$Z = -11.32$$

Reject H_0.

Conclusion: Americans are taller.

Single Proportion Test

Let

$$p = \frac{x}{n}$$

Population proportion

$$P$$

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

where

$$Q = 1 - P$$

Example

20 people attacked by disease

18 survived

$$p = 0.9$$

Test

$$H_0 : P = 0.85$$

$$Z = 0.6265$$

Accept H_0.

Example

1000 people sample

540 vegetarians

$$p = 0.54$$

Test

$$H_0 : P = 0.5$$

$$Z = 2.532$$

Since

$$2.532 < 2.58$$

Accept H_0.

Manufacturer Defective Products Example

Claim

$$P = 0.04$$

Sample

$$n = 500$$

Defective

$$x = 100$$

$$p = 0.20$$

$$Z = 18.26$$

Since

$$|Z| > 1.96$$

Reject H_0.

The claim is false.

Difference of Proportions

Test statistic

$$Z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where

$$P = \frac{x_1 + x_2}{n_1 + n_2}$$

Example

City A

$$p_1 = 0.20$$

$$n_1 = 900$$

City B

$$p_2 = 0.185$$

$$n_2 = 1600$$

$$Z = 0.92$$

Difference is not significant.

Example

Town voters

$$p_1 = 0.60$$

County voters

$$p_2 = 0.48$$

$$Z = 2.9$$

Reject H_0.

Town voters favor proposal more.

Example

University exam failures

Sample 1

$$n_1 = 400, \quad p_1 = 0.75$$

Sample 2

$$n_2 = 500, \quad p_2 = 0.60$$

$$Z = 4.74$$

Reject H_0.

Practice Problems

Exercise 1

70% customers satisfied

Sample

$$n = 200, \quad x = 150$$

Result

Fail to reject H_0.

Exercise 2

Drug success rate ≥60%

Sample

$$n = 100, \quad x = 55$$

Result

Fail to reject H_0.

Exercise 3

45% political support

Sample

$$n = 500, \quad x = 240$$

Result

Fail to reject H_0.

Exercise 4

Product popularity among men and women

Result

Reject H_0.

Exercise 5

Smoking proportion comparison

Result

Fail to reject H_0.